# TARGET PRODUCT PROFILE

## Naval Anti-Corrosive & Anti-Fouling Maritime Coating

*Amphora Labs · HII Collaboration*

| Document Type | Prepared By | Date Issued | Version |
|---|---|---|---|
| Target Product Profile | Amphora Labs | June 20, 2026 | v1.0 — HII Delivery Draft |
| **Program** | **Platform** | **Chemistry** | **Dev Phase** |
| MIL-DTL Maritime Coating | Surface Ships & Submarines | Epoxy-Polyamide, 2-Part | Q2–Q4 2026 (Weeks 1–24) |

## 1. PURPOSE & SCOPE

This Target Product Profile (TPP) defines the required, target, and ideal performance attributes for the Amphora Labs AI-designed naval maritime coating developed under the HII collaboration. The document serves as the primary technical specification governing formulation design, AI-model optimization constraints, laboratory testing, and MIL-DTL qualification submission.

The coating system is a two-part epoxy-polyamide architecture specifically engineered for hull and below-waterline applications on U.S. Navy surface vessels and submarines, including Virginia-class submarines and Ford-class carriers. All design decisions are constrained by the Amphora Manufacturing Compiler—ensuring every candidate formulation is synthesizable from commercially available, non-restricted feedstocks.

### Primary Performance Objectives

- Exceptional anti-corrosion resistance in high-salinity marine environments
- Anti-fouling performance to suppress biofouling (barnacles, algae, biofilm) without heavy metals
- Adhesion integrity on steel substrates across thermal cycling and dynamic loading
- VOC compliance per EPA Method 24 and applicable MIL-SPEC regulations
- Drop-in applicability using existing shipyard spray equipment—no retooling
- MIL-DTL qualification pathway within 24-week program timeline

## 2. TARGET PRODUCT PROFILE — PERFORMANCE SPECIFICATIONS

All property tiers are defined as: Minimum Acceptable (qualification floor), Target (AI optimization objective, MIL-SPEC compliant), and Ideal/Stretch (best-in-class fleet impact). The AI model will be ranked against the Target tier using the Synthesis Delta metric.

| Property | Min. Acceptable | Target (AI-Optimized) | Ideal / Stretch | Test Standard |
|---|---|---|---|---|
| **ANTI-CORROSION PERFORMANCE** | | | | |
| **Salt Spray Resistance** | ≥500 hrs, no blistering (ASTM B117) | ≥1,000 hrs, no corrosion creep >2 mm from scribe | ≥2,000 hrs with zero | ASTM B117 |

| Property | Min. Acceptable | Target (AI-Optimized) | Ideal / Stretch | Test Standard |
|---|---|---|---|---|
| | | | visible defects | |
| **Pitting Corrosion Potential** | >0.1 V vs. SCE | >0.2 V vs. SCE | >0.35 V vs. SCE | ASTM G5 / EIS |
| **Galvanic Compatibility (dissimilar metals, Al–Steel)** | <0.25 V potential difference | <0.15 V per MIL-STD-889 | <0.08 V | MIL-STD-889 |
| **Cathodic Disbondment Resistance** | ≤15 mm radius after 30 days | ≤8 mm radius after 30 days | ≤5 mm radius after 60 days | ASTM G8 / G95 |
| **Corrosion Underfilm Creep** | ≤3 mm from scribe | ≤1 mm from scribe | Zero creep | ASTM D1654 |
| **ANTI-FOULING PERFORMANCE** | | | | |
| **Biofouling Resistance (barnacles, mussels, algae)** | ≤15% surface coverage after 12-month immersion | ≤5% surface coverage; no hard foulers | Zero macrofouling after 18 months | ASTM D3623 / ISO 15181 |
| **Biofilm / Slime Prevention** | Reduction ≥50% vs. uncoated steel | Reduction ≥80% vs. control (ISO 19458) | Reduction ≥95% at 6 months | ISO 19458 / ASTM E2562 |
| **Frictional Drag Coefficient Increase** | <15% increase over service life vs. baseline | <8% increase at 24 months | <3% increase; net fuel savings measurable | ITTC-7.5-02-03 |
| **Biocide / Active Agent Profile** | Heavy-metal-free; compliant with 33 CFR 151 | Copper-free; non-toxic biocide package or fouling-release mechanism | Biocide-free (passive fouling-release only) | 33 CFR 151 / EPA FIFRA |
| **Leach Rate (if biocide-containing formulation)** | ≤5 µg/cm²/day | ≤2 µg/cm²/day | 0 µg/cm²/day (fouling-release only) | ISO 15181-2 |
| **MECHANICAL & ADHESION PROPERTIES** | | | | |
| **Adhesion Strength (pull-off, steel substrate)** | ≥5 MPa | ≥8 MPa | ≥12 MPa | ASTM D4541 |
| **Flexibility / Mandrel Bend** | Pass 6mm mandrel, no cracking | Pass 3mm mandrel | Pass 2mm mandrel | ASTM D522 |
| **Impact Resistance** | Pass 80 in-lb direct impact | Pass 120 in-lb direct impact | Pass 160 in-lb | ASTM D2794 |
| **Abrasion Resistance (Taber CS-17, 1000 cycles)** | ≤150 mg weight loss | ≤80 mg weight loss | ≤50 mg weight loss | ASTM D4060 |
| **Glass Transition Temperature (Tg)** | >120°C | >150°C | >170°C | ASTM E1356 (DSC) |
| **APPLICATION & MANUFACTURING CONSTRAINTS** | | | | |

| Property | Min. Acceptable | Target (AI-Optimized) | Ideal / Stretch | Test Standard |
|---|---|---|---|---|
| **Mix Ratio (Volume)** | 1:1 ± 0.15 (Resin:Hardener) | 1:1 ± 0.10 exact (AHEW×ρA)/(EEW×ρB) | 1:1 ± 0.05 | Internal / ASTM D1475 |
| **Viscosity at 25°C (mixed)** | 100–600 cP | 200–500 cP (spray-optimized) | 250–400 cP | ASTM D2196 |
| **Pot Life at 25°C** | ≥1.5 hours | 1.5–4 hours | 2–3 hours (optimal open time) | ASTM D1200 |
| **Dry Film Thickness (DFT)** | 100–500 µm | 125–375 µm per coat | 150–250 µm single-coat | ASTM D7091 / SSPC-PA 2 |
| **Surface Prep Requirement** | SSPC-SP6 (Commercial Blast) | SSPC-SP10 (Near-White Blast) | SSPC-SP5 (White Metal) for extreme service | SSPC-SP10 |
| **VOC Content** | ≤340 g/L (EPA Method 24) | ≤340 g/L | ≤250 g/L (AQMD compliant) | EPA Method 24 |
| **THERMAL & CHEMICAL RESISTANCE** | | | | |
| **Thermal Cycling Resistance** | No delamination: -40°C to +80°C (50 cycles) | No defects: -51°C to +120°C (100 cycles) | Retain >95% adhesion over 200 cycles | ASTM D2485 / MIL-C-46168 |
| **Seawater Immersion Resistance** | Pass 1,000 hrs continuous immersion (ASTM D870) | Pass 2,000 hrs; adhesion ≥80% retained | Pass 5,000 hrs; no blistering | ASTM D870 |
| **Fuel & Solvent Spot Resistance** | No softening after 24-hr exposure (JP-5, Diesel) | No blistering, softening, or adhesion loss at 48 hrs | Full chemical resistance to JP-8, AFFF, seawater | ASTM D543 |
| **UV / Weathering Resistance (topside supplement)** | Pass 500 hrs QUV, gloss retention ≥50% | Pass 1,000 hrs QUV, gloss retention ≥70% | Pass 2,000 hrs; ΔE <2.0 | ASTM G154 |
| **QUALIFICATION & REGULATORY COMPLIANCE** | | | | |
| **Primary MIL-SPEC Target** | MIL-DTL-24441 (Type III or IV equivalent) | Full MIL-DTL-24441 qualification submission | Dual qualification: MIL-DTL-24441 + MIL-PRF-23236 | MIL-DTL-24441 |
| **Anti-Fouling Regulatory** | 33 CFR Part 151 compliant | IMO AFS Convention compliant (TBT-free) | EPA Design for the Environment (DfE) certified | 33 CFR 151 / IMO AFS |

| Property | Min. Acceptable | Target (AI-Optimized) | Ideal / Stretch | Test Standard |
|---|---|---|---|---|
| **NAVSEA Approval** | Compatible with NAVSEA Standard Items | NAVSEA Standard Item 009-32 compliant | NAVSEA-approved QPL listing | NAVSEA 0900-LP-001-7000 |
| **Shelf Life** | ≥12 months (unmixed components) | ≥18 months at 15–35°C | ≥24 months with no property degradation | ASTM D1475 |

## 3. AI MODEL GENERATION STRATEGY & MANUFACTURING CONSTRAINTS

The Amphora Manufacturing Compiler operates under a Manufacturing-First constraint paradigm—every candidate coating formulation must be synthesizable before it is optimized for performance. The AI model does not generate candidates and then filter; it constrains first, then generates.

### Hard Constraints (Binary Pass/Fail — Manufacturing Firewall)

- VOC Compliance: ≤340 g/L enforced deterministically via EPA Method 24 logic
- Stoichiometric Match: (AHEW × ρA) / (EEW × ρB) = 1.0 ± 0.10 — 1:1 volume ratio verified before any candidate advances
- Equipment Fit: Viscosity < 500 cP at 25°C; Pot Life 1.5–4 hours (compatible with existing shipyard spray equipment)
- Feedstock Availability: All monomers sourced from Tier 1 (Sigma-Aldrich, TCI) or Tier 2 (HII-specified precursors)—no restricted or scarce elements
- Synthesis Feasibility: Maximum 2-step synthesis pathway (exception: up to 4 steps if Tg gain >15%—open decision item)

### Optimization Objective (Synthesis Delta)

Candidates are ranked by the Synthesis Delta: a weighted heuristic combining performance scores against the Target tier above and precursor cost/availability. Top 5–10 candidates will be delivered to HII for physical lab testing in Q3 2026.

### Feedstock Seed Library

- Tier 1: ~100 commercial monomers (off-the-shelf, real-time cost/availability ingested)
- Tier 2: HII-specific precursors provided by partner
- Exclusion: Cobalt and other scarce/restricted strategic metals

## 4. DEVELOPMENT MILESTONES & PROGRAM STATUS

| Phase | Timeline | Deliverable | HII Outcome | Status |
|---|---|---|---|---|
| **1** | Weeks 1–6 (Q2 2026) | Manufacturing Firewall deployed; VOC & stoichiometry constraint engine active; Feedstock Seed Library digitized (~100 monomers) | 100% elimination of unbuildable chemistry leads | **COMPLETE** |

| Phase | Timeline | Deliverable | HII Outcome | Status |
|---|---|---|---|---|
| 2 | Weeks 7–12 (Q3 2026) | Seeded Generation Engine (GraphINVENT / DDPM); Dual-graph Resin–Hardener matching; 1:1 volume stoichiometry optimization | Resin–Hardener candidate pairs with verified curing kinetics | IN PROGRESS |
| 3 | Weeks 13–18 (Q3 2026) | ASTM Judge Bank trained (B117 salt fog, D4541 adhesion); Synthesis Delta ranking of top candidates by Tg >150°C and supply chain stability | 5–10 validated formulations delivered to HII for physical lab testing | PENDING |
| 4 | Weeks 19–24 (Q4 2026) | Physical synthesis and MIL-SPEC testing; DSC/TGA/Adhesion data fed back into NSKG; HII ROI & Qualification Report prepared | Final maritime coating candidates submitted for MIL-DTL qualification | PENDING |

## 5. OPEN ITEMS & DECISION LOG

| # | Open Item | Owner | Priority |
|---|---|---|---|
| OQ-1 | Synthesis step limit: hard-code 2-step maximum or allow up to 4 steps when Tg gain >15%? | Amphora / HII SME | HIGH |
| OQ-2 | ASTM B117 salt spray test duration confirmation: 500 hrs vs. 1,000 hrs? (Awaiting HII clarification) | HII | HIGH |
| OQ-3 | Anti-fouling mechanism preference: biocide-based vs. fouling-release polymer — HII environmental policy and theater-of-operation constraints required | HII / NAVSEA | HIGH |
| OQ-4 | MIL-DTL target qualification tier: Type III vs. Type IV, and whether dual qualification (MIL-PRF-23236) is in scope for this program phase | HII Program Office | MEDIUM |
| OQ-5 | Topside vs. below-waterline scope boundary: Does this TPP cover boot-top and topsides, or hull/boot-top only? UV resistance tier is contingent. | Amphora / HII | MEDIUM |
| OQ-6 | HII Tier 2 precursor list: Required for Feedstock Seed Library finalization. Provision of proprietary precursor data under NDA needed. | HII | HIGH |

## 6. APPROVALS & DOCUMENT CONTROL

This document requires sign-off from both Amphora Labs and HII before use as a binding specification for AI model optimization and laboratory testing.

| Role | Name | Signature | Date |
|---|---|---|---|
| **Amphora Labs — Principal Investigator** | | | |

| Role | Name | Signature | Date |
|---|---|---|---|
| **Amphora Labs — AI / Materials Science Lead** | | | |
| **HII — Program Manager** | | | |
| **HII — Materials & Coatings SME** | | | |
| **NAVSEA Technical Representative (optional)** | | | |

*Next scheduled TPP review: Upon completion of Phase 2 (Weeks 7–12) or upon resolution of high-priority open items.*